LaTeX Preamble & Setup Quick Lookup

Packages, settings, shortcuts, and practical tricks before `\begin{document}`

For beginning LaTeX users in coursework and research

The focus of this handbook is the **setup area before writing content**. Optional lines are often shown commented with `%`; remove the leading `%` only when you need that feature.

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1. How to use this guide if you know almost no LaTeX

LaTeX has two broad parts:

1. the **preamble**: everything before `\begin{document}`; this is where you load packages and define global rules;
2. the **document body**: everything between `\begin{document}` and `\end{document}`; this is where you write the paper.

A useful mental model is:

`document class` → `packages` → `global settings/macros` → `\begin{document}` → `paper content`.

1.1 A tiny file that compiles

```
\documentclass[12pt]{article}
\usepackage[margin=1in]{geometry}
\usepackage{amsmath}

\title{My First Economics Note}
\author{Your Name}
\date{\today}

\begin{document}
\maketitle

This is ordinary text. Inline math looks like $u(c)$.

\begin{equation}
U = \sum_{t=0}^{T} \beta^t u(c_t).
\end{equation}

\end{document}
```

What happens when you compile it? LaTeX reads the setup first, then creates the title, text, and numbered equation. The source file is `.tex`; the usual finished output is a PDF.

1.2 The three questions to ask before adding anything

Before adding a package or setting, ask:

(1) **Do I actually need it?** Keep the preamble small while learning.
(2) **Does my class/journal template already provide it?** Avoid fighting an official template.
(3) **Is this global or local?** Global formatting belongs in the preamble; one-off formatting usually belongs in the body.

Tip. For a new project, first make the smallest file compile. Add figures, bibliography, custom macros, and special formatting one module at a time.

2. Compiler / engine choice: pdfLaTeX, XeLaTeX, or LuaLaTeX

For a beginner writing an English-language academic paper, **pdfLaTeX is usually the easiest default**. XeLaTeX and LuaLaTeX are especially useful when you need system fonts or multilingual text.

Engine	Good default for	Typical font setup
pdfLaTeX	Coursework, papers, journal-style PDFs	Traditional LaTeX font packages such as `newtxtext/newtxmath`
XeLaTeX	System fonts, Unicode-heavy documents, multilingual work	`fontspec`
LuaLaTeX	Similar benefits to XeLaTeX plus Lua-based features	`fontspec`

For modern pdfLaTeX installations, UTF-8 input is normally the default. You may still encounter older templates containing:

```
% Mostly needed in older pdfLaTeX templates:
% \usepackage[utf8]{inputenc}
% \usepackage[T1]{fontenc}
```

For XeLaTeX/LuaLaTeX, a basic system-font setup looks like:

```
% Compile with XeLaTeX or LuaLaTeX -- not pdfLaTeX
% \usepackage{fontspec}
% \setmainfont{Times New Roman}
```

Watch out. Do not combine `fontspec` with pdfLaTeX. If a template already chooses an engine/font system, follow the template.

3. The 30-second starter preamble

For most quantitative or technical homework, start here. It is intentionally small.

```
\documentclass[12pt]{article}

% ----- Math -----
\usepackage{amsmath}     % align, cases, equation tools
\usepackage{amssymb}     % extra math symbols
\usepackage{amsfonts}    % \mathbb{R}, \mathbb{E}, etc.
\usepackage{mathtools}   % useful extensions to amsmath
\usepackage{bm}          % bold math: \bm{\beta}, \bm{x}

% ----- Page layout -----
\usepackage[margin=1in]{geometry}

% ----- Useful for economics tables -----
\usepackage{booktabs}

% ----- Hyperlinks / references -----
\usepackage[hidelinks]{hyperref}

\title{ECONS XXX -- Homework X}
\author{Your Name}
\date{\today}

\begin{document}
% Your content begins here.
\end{document}
```

Why. This covers clean math, standard margins, tables, and clickable PDF references without loading a large number of packages.

Tip. Keep a master copy. Add optional modules only when needed instead of starting from a giant preamble every time.

4. How comments work in the setup area

Anything after `%` on a line is ignored by LaTeX. This makes comments perfect for keeping optional packages and reminders in your template.

```
% OPTIONAL: remove the % when needed
% \usepackage{graphicx}      % figures
% \usepackage{float}         % [H] figure placement
```

```
% \usepackage{subcaption}    % subfigures
% \usepackage{setspace}      % line spacing
% \usepackage{fancyhdr}      % headers / footers
% \usepackage{siunitx}       % decimal alignment in tables
```

To activate a package, change

```
% \usepackage{graphicx}
```

to

```
\usepackage{graphicx}
```

Watch out. Do not uncomment every package just because it is available. More packages can mean more conflicts and slower debugging.

5. Document class and global options

The first line controls the overall document type.

```
\documentclass[12pt]{article}

% Alternatives:
% \documentclass[11pt]{article}       % slightly more compact
% \documentclass[12pt]{report}        % chapters; longer reports
% \documentclass[12pt]{book}          % dissertation / book style

% Useful options:
% \documentclass[12pt,fleqn]{article} % display equations left-aligned
% \documentclass[12pt,leqno]{article} % equation numbers on the left
```

For most problem sets and research papers, `article` is usually the simplest choice.

6. Page layout, margins, and paper size

6.1 Geometry

```
\usepackage{geometry}
\geometry{margin=1in}

% North America:
% \geometry{letterpaper, margin=1in}

% Many international settings:
% \geometry{a4paper, margin=2.5cm}

% Fine control:
% \geometry{
%   top=1in,
%   bottom=1in,
%   left=1.1in,
%   right=1.1in
% }
```

Tip. If an instructor or journal specifies margins, set them once here. Do not manually push text with repeated `hspace` or
`vspace`.

6.2 Useful geometry patterns

You can change only the dimension you need:

```
% Wider left margin for binding / notes:
% \geometry{left=1.25in, right=1in, top=1in, bottom=1in}

% Add room between the header and body:
% \geometry{headsep=18pt}

% Include header/footer in the measured page area:
```

```
% \geometry{includeheadfoot, margin=1in}
```

Why. A page-layout package is more stable than scattering manual spaces throughout a paper. If the paper later changes from 1-inch to 1.25-inch margins, one preamble edit updates the entire document.

6.3 Paragraph style

```
% Default LaTeX: indented paragraphs, little vertical gap.

% Optional "block paragraph" style:
% \setlength{\parindent}{0pt}
% \setlength{\parskip}{6pt}
```

For formal papers, the default paragraph indentation is often safer unless a style guide says otherwise.

7. Title, authors, affiliations, acknowledgments, and PDF metadata

The simplest setup is:

```
\title{The Economics of Something Important}
\author{Your Name}
\date{September 2026}

% In the body:
% \maketitle
```

A common paper-style pattern uses `\thanks` for affiliations or acknowledgments:

```
\title{Paper Title}
\author{
  First Author\thanks{Department of Economics, University A. Email: a@uni.edu.}
  \and
  Second Author\thanks{Department of Economics, University B. Email: b@uni.edu.}
}
\date{\today}
```

You can also set PDF metadata through `hyperref`:

```
\usepackage[hidelinks]{hyperref}
\hypersetup{
  pdftitle={Paper Title},
  pdfauthor={Your Name},
  pdfsubject={Economics},
  pdfkeywords={labor economics, productivity, firms}
}
```

Tip. Do not manually type a title at the top of page 1 unless a template requires it. Defining `\title`, `\author`, and `\date` keeps the source reusable.

8. Fonts and typography

8.1 Keep the default, or choose one family

```
% Default LaTeX font is already suitable for academic work.

% Times-like text + math:
% \usepackage{newtxtext}
% \usepackage{newtxmath}

% Palatino-like text + math:
% \usepackage{newpxtext}
% \usepackage{newpxmath}

% Latin Modern:
% \usepackage{lmodern}
```

Watch out. Do not load several competing text/math font packages together. Pick one font system.

8.2 Microtype: an easy polish improvement

```
% Recommended for polished PDF typography:
% \usepackage{microtype}
```

It improves line breaking, spacing, and character protrusion with almost no effort.

9. Line spacing and paragraph spacing

```
% \usepackage{setspace}

% \onehalfspacing     % 1.5 line spacing
% \doublespacing      % double spacing
% \singlespacing      % single spacing

% Local spacing only:
% \begin{spacing}{1.2}
% Text here...
% \end{spacing}
```

Tip. Use these settings in the preamble if your instructor requires 1.5 or double spacing. Avoid manually inserting blank lines everywhere.

10. Core math packages for technical writing

A practical math block is:

```
\usepackage{amsmath}     % align, split, cases, equation environments
\usepackage{amssymb}     % symbols such as \mathbb and many relations
\usepackage{amsfonts}    % math fonts
\usepackage{mathtools}   % extensions and fixes for amsmath
\usepackage{bm}          % bold vectors and Greek symbols

% Optional:
% \usepackage{cancel}    % show cancellation in derivations
% \usepackage{physics}   % convenient notation, but can redefine commands
```

Typical uses:

Package	Why researchers often use it
`amsmath`	Multi-line derivations, FOCs, systems, equation numbering
`amssymb / amsfonts`	Blackboard bold sets and many symbols
`mathtools`	Better delimiters, math extensions, small fixes
`bm`	Bold vectors such as `\bm{x}` and `\bm{beta}`

10.1 What these packages let you write

Once the math packages are loaded, ordinary quantitative notation becomes straightforward. For example:

```
\[
\E[Y \mid X=x], \qquad
\bm{\beta} = (\beta_1,\ldots,\beta_K)', \qquad
A \in \mathbb{R}^{n\times k}.
\]
```

In the compiled PDF, the intended mathematical structure is visible as

$$\mathbb{E}[Y \mid X = x], \qquad \boldsymbol{\beta} = (\beta_1, \ldots, \beta_K)', \qquad A \in \mathbb{R}^{n \times k}.$$

Here `\qquad` is simply a convenient large horizontal math space; it is not part of the mathematics itself.

11. Create your own notation shortcuts in the preamble

Repeated notation becomes easier to read and maintain if you define it once.

```
% ---------- Probability and sets ----------
```

```
% \newcommand{\E}{\mathbb{E}}
% \newcommand{\Prob}{\mathbb{P}}
% \newcommand{\R}{\mathbb{R}}
% \newcommand{\N}{\mathbb{N}}

% ---------- Operators ----------
% \DeclareMathOperator{\Var}{Var}
% \DeclareMathOperator{\Cov}{Cov}
% \DeclareMathOperator{\Corr}{Corr}
% \DeclareMathOperator{\plim}{plim}
% \DeclareMathOperator*{\argmax}{arg\,max}
% \DeclareMathOperator*{\argmin}{arg\,min}

% ---------- Differential ----------
% \newcommand{\dd}{\,\mathrm{d}}

% ---------- Common bold notation ----------
% \newcommand{\bx}{\bm{x}}
% \newcommand{\by}{\bm{y}}
% \newcommand{\bX}{\bm{X}}
% \newcommand{\bbeta}{\bm{\beta}}
% \newcommand{\beps}{\bm{\varepsilon}}
```

After activation, you can write concise source such as

```
% \E[Y \mid X], \Var(X), X \in \R
% \plim \hat{\beta}, \argmax_x U(x)
```

Why. If you later change notation, you edit one definition rather than hundreds of equations.

11.1 Commands with arguments

A macro can accept input. This is useful for notation that has a repeated structure:

```
% One argument:
% \newcommand{\Econd}[1]{\mathbb{E}\!\left[#1\right]}
%
% Body:
% \Econd{Y_i \mid X_i}
```

which produces the same style every time:

$$\mathbb{E}[Y_i \mid X_i]\,.$$

With two arguments:

```
% \newcommand{\deriv}[2]{\frac{\partial #1}{\partial #2}}
% Body: \deriv{U}{c}
```

11.2 Paired delimiters with mathtools

For absolute values and norms that can resize automatically:

```
% Requires mathtools
% \DeclarePairedDelimiter{\abs}{\lvert}{\rvert}
% \DeclarePairedDelimiter{\norm}{\lVert}{\rVert}
%
% \abs{x}
% \abs*{\frac{x}{y}}     % starred form auto-sizes
% \norm*{\bm{\beta}}
```

Watch out. Avoid redefining a very common built-in command unless you know exactly what it does. Choose descriptive macro names and keep all your custom definitions in one block.

12. Equation numbering and reference setup

```
% Number equations by section:
% \numberwithin{equation}{section}

% Example labels in the document body:
```

```
% \begin{equation}
% y_i = \beta_0 + \beta_1 x_i + \varepsilon_i
% \label{eq:regression}
% \end{equation}

% Refer to it with:
% Equation~\eqref{eq:regression}
```

Recommended label prefixes:

`eq:`	equation	`fig:`	figure
`tab:`	table	`sec:`	section
`app:`	appendix	`thm:`	theorem

Tip. Put `label{...}` immediately after `caption{...}` in figures and tables.

12.1 A complete equation-reference example

```
\begin{equation}
  y_i = \alpha + \beta x_i + \varepsilon_i,
  \label{eq:baseline}
\end{equation}

As shown in Equation~\eqref{eq:baseline}, ...
```

The label is a stable internal name. If the equation moves from (2) to (5), the reference updates automatically after recompilation.

For multi-line mathematics:

```
\begin{align}
U(c_1,c_2) &= u(c_1) + \beta u(c_2), \\
\text{s.t.}\qquad c_1 + \frac{c_2}{1+r} &= y_1 + \frac{y_2}{1+r}.
\end{align}
```

Tip. Put `&` immediately before the symbol you want aligned, commonly `=` or `\le`. Use `\notag` on a line that should not receive an equation number.

13. Hyperlinks and smart references

```
% Usually load near the end of the preamble:
\usepackage[hidelinks]{hyperref}

% Optional smart references -- load AFTER hyperref:
% \usepackage[nameinlink,noabbrev]{cleveref}
```

With `cleveref`, source can become:

```
% \cref{fig:eventstudy}
% \cref{tab:mainreg}
% \cref{eq:FOC}
```

instead of manually typing “Figure”, “Table”, or “Equation” each time.

13.1 The nonbreaking-space trick

```
Figure~\ref{fig:main}
Table~\ref{tab:main}
Section~\ref{sec:data}
Equation~\eqref{eq:model}
```

The `~` prevents the label word and its number from splitting across lines.

13.2 Manual vs. automatic reference names

Source	Typical output / idea
`Figure~\ref{fig:main}`	You supply the word "Figure" yourself.
`\autoref{fig:main}`	`hyperref` supplies the object name.
`\cref{fig:main}`	`cleveref` supplies the object name and handles multiple labels well.

Tip. For a large paper with many tables, figures, equations, appendices, and propositions, `cleveref` can substantially reduce manual reference formatting.

14. Tables: a professional research-paper setup

14.1 Packages worth keeping available

```
\usepackage{booktabs}        % professional horizontal rules

% \usepackage{array}         % custom column types
% \usepackage{tabularx}      % tables fit to a target width
% \usepackage{longtable}     % tables spanning multiple pages
% \usepackage{multirow}      % cells spanning rows
% \usepackage{siunitx}       % decimal alignment / number formatting
% \usepackage{threeparttable}% table notes
```

Why. Regression and summary-statistics tables are common in quantitative research papers. `booktabs` gives clean
`toprule`,
`midrule`, and
`bottomrule`.

14.2 Decimal alignment with siunitx

```
% \usepackage{siunitx}
%
% \begin{tabular}{l S[table-format=2.2]}
% \toprule
% Variable & {Mean} \\
% \midrule
% Income   & 12.34 \\
% Age      & 30.10 \\
% \bottomrule
% \end{tabular}
```

Tip. For professional academic tables, avoid vertical lines unless specifically required.

14.3 A complete regression-style table pattern

This is the body code you are preparing the preamble to support:

```
\begin{table}[htbp]
\centering
\caption{Summary Statistics}
\label{tab:summary}
\begin{threeparttable}
\begin{tabular}{l S[table-format=4.1] S[table-format=3.2] S[table-format=6.0]}
\toprule
Variable & {Mean} & {SD} & {N} \\
\midrule
Income   & 4250.5 & 811.32 & 12500 \\
Age      &   38.7 &  12.41 & 12500 \\
\bottomrule
\end{tabular}
\begin{tablenotes}[flushleft]\footnotesize
\item Notes: This is a toy example showing table structure.
\end{tablenotes}
\end{threeparttable}
\end{table}
```

The central pieces are `booktabs` for clean rules, `siunitx` for numeric alignment, and `threeparttable` for notes that

belong to the table.

Compiled miniature example

Table 1: *

Variable	Mean	SD	N
Income	4250.5	811.32	12 500
Age	38.7	12.41	12 500

Notes: Numbers align cleanly without vertical rules.

14.4 Useful siunitx setup for regression tables

```
% In the preamble:
% \sisetup{
%   detect-weight=true,
%   detect-inline-weight=math,
%   table-number-alignment=center
% }
```

Watch out. If your regression table is generated by Stata/R, do not manually retype hundreds of coefficients into LaTeX. Export a `.tex` fragment and include it with `\input`.

15. Figures, plots, and subfigures

```
% \usepackage{graphicx}     % \includegraphics
% \usepackage{float}        % enables [H]
% \usepackage{caption}      % caption control
% \usepackage{subcaption}   % subfigures
```

A useful folder shortcut:

```
% Requires graphicx
% \graphicspath{{figures/}}
```

Then you can write `\includegraphics{eventstudy.pdf}` instead of the full folder path.

Common float placement letters:

Option	Meaning
`h`	approximately here
`t`	top of page
`b`	bottom of page
`p`	dedicated float page
`htbp`	let LaTeX choose among sensible locations
`H`	exactly here; requires `float`

Watch out. Do not use `[H]` for every figure. `[htbp]` usually produces a better paper layout.

15.1 A complete figure pattern

```
\begin{figure}[htbp]
  \centering
  \includegraphics[width=0.75\linewidth]{figures/main_result.pdf}
  \caption{Main Result}
  \label{fig:main}
\end{figure}

See Figure~\ref{fig:main}.
```

Why `\linewidth`? Inside ordinary text it is usually the available line width; inside a smaller container such as a `minipage`, it automatically adapts to that container. This is often safer than hard-coding inches.

Common sizing patterns:

```
% Fill available local width:
% \includegraphics[width=\linewidth]{figure.pdf}

% Use 70% of available local width:
% \includegraphics[width=.70\linewidth]{figure.pdf}

% Limit height while preserving aspect ratio:
% \includegraphics[height=.55\textheight,keepaspectratio]{figure.pdf}

% Crop whitespace: trim = left bottom right top
% \includegraphics[width=.8\linewidth,trim=10 5 10 5,clip]{figure.pdf}
```

15.2 Subfigures / panels

```
% Preamble:
% \usepackage{subcaption}
%
% Body:
% \begin{figure}[htbp]
% \centering
% \begin{subfigure}{.48\linewidth}
%   \includegraphics[width=\linewidth]{panel_a.pdf}
%   \caption{Panel A}
%   \label{fig:panel-a}
% \end{subfigure}\hfill
% \begin{subfigure}{.48\linewidth}
%   \includegraphics[width=\linewidth]{panel_b.pdf}
%   \caption{Panel B}
%   \label{fig:panel-b}
% \end{subfigure}
% \caption{Two Related Outcomes}
% \label{fig:two-panels}
% \end{figure}
```

16. Control floating tables and figures

If LaTeX moves a table too far away from the relevant section:

```
% \usepackage[section]{placeins}
```

This inserts barriers at sections. Or use a manual barrier:

```
% \FloatBarrier
```

17. Bibliography and citation setup

Choose **one** main workflow.

17.1 Option A: biblatex + biber

```
% \usepackage{csquotes}
% \usepackage[
%   backend=biber,
%   style=authoryear,
%   natbib=true,
%   maxcitenames=2,
%   maxbibnames=99
% ]{biblatex}
%
% \addbibresource{references.bib}
```

Common body commands:

```
% \textcite{angrist2009}     % Angrist and Pischke (2009)
% \parencite{angrist2009}    % (Angrist and Pischke, 2009)
% \printbibliography
```

17.2 Option B: natbib + BibTeX

```
% \usepackage[round,authoryear]{natbib}
%
% Body:
% \citet{angrist2009}        % Angrist and Pischke (2009)
% \citep{angrist2009}        % (Angrist and Pischke, 2009)
%
% End of document:
% \bibliographystyle{apalike}
% \bibliography{references}
```

Watch out. Do not independently configure both `biblatex` and `natbib` in the same document. Choose the workflow required by your class, coauthors, or journal.

17.3 What goes inside references.bib?

A typical article entry looks like:

```
@article{acemoglu2001,
  author  = {Daron Acemoglu and Joshua Angrist},
  title   = {Consequences of Employment Protection?},
  journal = {Journal of Political Economy},
  year    = {2001},
  volume  = {109},
  number  = {5},
  pages   = {915--957}
}
```

The key `acemoglu2001` is the identifier used by your citation command. For example:

```
\citet{acemoglu2001}
\citep{acemoglu2001}
% or, with biblatex:
\textcite{acemoglu2001}
```

17.4 Compile sequence: why the bibliography sometimes appears only after several runs

With BibTeX, the conceptual sequence is:

```
pdflatex main
bibtex main
pdflatex main
pdflatex main
```

With `biblatex` + Biber:

```
pdflatex main
biber main
pdflatex main
pdflatex main
```

If you use `latexmk`, it can usually determine the required repeated passes automatically:

```
latexmk -pdf main.tex
```

Tip. In Overleaf, the editor usually handles the repeated passes for you. Locally, `latexmk` is one of the easiest ways to avoid thinking about the exact compilation sequence.

18. Theory papers: theorem, proposition, assumption

```
% \usepackage{amsthm}
%
% \newtheorem{theorem}{Theorem}
% \newtheorem{proposition}{Proposition}
% \newtheorem{lemma}{Lemma}
% \newtheorem{corollary}{Corollary}
%
% \theoremstyle{definition}
```

```
% \newtheorem{definition}{Definition}
% \newtheorem{assumption}{Assumption}
%
% \theoremstyle{remark}
% \newtheorem{remark}{Remark}
```

Number by section if desired:

```
% \newtheorem{theorem}{Theorem}[section]
```

A body example:

```
\begin{proposition}\label{prop:monotonicity}
If $u'(c)>0$, utility is increasing in consumption.
\end{proposition}

\begin{proof}
The result follows directly from the sign of the derivative.
\end{proof}
```

The point of the preamble declarations is that propositions, assumptions, lemmas, and proofs all receive consistent formatting and numbering throughout the paper.

19. Headers, footers, and page numbering

```
% \usepackage{fancyhdr}
%
% \pagestyle{fancy}
% \fancyhf{}                % clear existing header/footer fields
% \lhead{ECONS 5XX}
% \chead{Homework 3}
% \rhead{Your Name}
% \cfoot{\thepage}
%
% \renewcommand{\headrulewidth}{0pt}
% \renewcommand{\footrulewidth}{0pt}
```

Simpler alternatives:

```
% \pagestyle{plain}         % page number only
% \pagestyle{empty}         % no page numbers
```

20. Section-heading control

```
% \usepackage{titlesec}
%
% Compact spacing:
% \titlespacing*{\section}{0pt}{1.5ex}{0.8ex}
%
% Custom style:
% \titleformat{\section}
%   {\large\bfseries}
%   {\thesection}{1em}{}
```

Watch out. Journal templates often control headings themselves. If you are using an official class file, avoid overriding its style unless permitted.

21. Lists and compact outlines

```
% \usepackage{enumitem}
%
% Compact list defaults:
% \setlist[itemize]{noitemsep, topsep=2pt}
% \setlist[enumerate]{noitemsep, topsep=2pt}
%
% Lettered list:
% \begin{enumerate}[label=(\alph*)]
```

```
%   \item First item
% \end{enumerate}
```

Useful for assumptions, model steps, homework subparts, and referee-response style documents.

22. Appendix and landscape pages

22.1 Appendix

No package is required for the basic approach:

```
% In the document body:
% \appendix
% \section{Additional Results}
```

Optional enhancements:

```
% \usepackage[toc,page]{appendix}
```

22.2 Wide regression tables

```
% \usepackage{pdflscape}
%
% \begin{landscape}
%   ... very wide table ...
% \end{landscape}
```

This is better than manually rotating every cell.

23. Stata, R, Python, and external regression tables

For code snippets:

```
% \usepackage{listings}
```

For tables exported by Stata/R/Python, keep generated tables in separate files:

```
% Main paper:
% \input{tables/regression_main.tex}
% \input{tables/summary_statistics.tex}
```

Recommended project structure:

```
project/
  main.tex
  references.bib
  sections/
    introduction.tex
    model.tex
    data.tex
    results.tex
  figures/
    eventstudy.pdf
    trends.pdf
  tables/
    summary_statistics.tex
    regression_main.tex
```

Then assemble the paper cleanly:

```
% \input{sections/introduction}
% \input{sections/model}
% \input{sections/data}
% \input{sections/results}
```

Why. Separating tables, figures, and sections keeps a research project much easier to maintain and reproduce.

24. Small spacing and alignment commands worth recognizing

These are mostly body-level commands, but they are worth knowing because they often appear in examples and custom macros in the preamble.

Command	What it does	Typical use
`\,`	tiny math space	$f(x)\,\mathrm{d}x$
`\;`	larger math space	separate nearby math objects
`\!`	negative math space	tighten a wide construction
`\quad`	large fixed math space	separate equation and condition
`\qquad`	about twice `\quad`	$x = 1 \qquad i = 1, \ldots, n$
`~`	nonbreaking text space	`Table~\ref{...}`
`\hfill`	stretchable horizontal space	push material to opposite sides
`\noindent`	suppress one paragraph indent	special front matter / notes
`&`	alignment point in tables/`align`	align equations at equals signs
`\\`	end row/line in structured environments	tables and `align`

A classic quantitative-writing use of `\qquad` is:

```
\[
y_i = x_i'\beta + \varepsilon_i,
\qquad i=1,\ldots,n.
\]
```

which appears as

$$y_i = x_i'\beta + \varepsilon_i, \qquad i = 1, \ldots, n.$$

Watch out. Spacing commands should improve readability, not manually "draw" the page. Use semantic structures such as `align`, `tabular`, `minipage`, and page-layout packages before resorting to repeated manual spacing.

25. Draft-mode tricks that save time

25.1 Show overfull lines

```
% Draw a black marker where text exceeds the margin:
% \overfullrule=5pt
```

Remove it before final submission.

25.2 Show label names

```
% \usepackage{showkeys}
```

This displays label keys such as `eq:ols` or `fig:eventstudy` while drafting.

25.3 TODO notes

```
% \usepackage[colorinlistoftodos]{todonotes}
%
% Body examples:
% \todo{Check this result.}
% \todo[inline]{Rewrite this argument.}
%
% Final version switch:
% \usepackage[disable]{todonotes}
```

25.4 One draft/final switch

```
% \newif\ifdraft
% \drafttrue
% % \draftfalse
%
% \ifdraft
%   % draft-only packages/settings here
```

```
% \fi
```

Tip. A single switch is much easier than manually deleting draft annotations throughout a long paper.

26. Common compilation and debugging tricks

- If references show `??`, compile again. Cross-references generally need multiple passes.
- If a bibliography is missing, make sure you are using the correct BibTeX/Biber workflow and compile sequence.
- A missing closing brace `}` may cause an error many lines after the true mistake.
- If a figure is missing, first check the filename, extension, relative path, and whether `graphicx` is loaded.
- If a command is undefined, check which package provides it.
- If a package conflict appears, simplify the preamble and reintroduce optional packages one at a time.

Useful source-display trick:

```
% Show raw LaTeX inline:
% \verb|\frac{a}{b}|
%
% Show a raw block:
% \begin{verbatim}
% \frac{a}{b}
% \end{verbatim}
```

26.1 Common error messages: what to inspect first

Message / symptom	First checks
`Undefined control sequence`	Misspelling? Missing package? Command used with wrong engine?
`Missing $ inserted`	Math command outside math mode, or unmatched `$`.
`Extra alignment tab has been changed to \cr`	Too many `&` symbols for the table/alignment structure.
`Missing } inserted`	Unbalanced braces; inspect a few lines before the reported location.
`Runaway argument?`	Usually a missing closing brace or environment end.
`Reference shows ??`	Compile again; confirm the label exists and is unique.
`Citation shows ?`	Check bibliography key and BibTeX/Biber step.
`Overfull \hbox`	A line cannot break; inspect long URLs, code, math, or fixed-width content.
`Float(s) lost / figures far away`	Check float environments, placement options, barriers, and oversized figures.

26.2 A practical local compile command

If you have a TeX distribution installed locally:

```
latexmk -pdf main.tex
```

To clean auxiliary files later:

```
latexmk -c
```

27. Package order: a practical rule

A generally safe order is:

1. language / encoding,
2. fonts,
3. mathematics,
4. geometry and spacing,
5. tables and figures,
6. bibliography,
7. colors,
8. `hyperref`,
9. `cleveref`.

Example skeleton:

```
\documentclass[12pt]{article}

% 1. Language / font
% \usepackage[english]{babel}
% \usepackage{newtxtext,newtxmath}

% 2. Math
\usepackage{amsmath,amssymb,amsfonts,mathtools,bm}

% 3. Layout
\usepackage[margin=1in]{geometry}
% \usepackage{setspace}

% 4. Tables / figures
\usepackage{booktabs}
% \usepackage{graphicx}
% \usepackage{siunitx}

% 5. Bibliography
% ... choose biblatex OR natbib ...

% 6. Links near the end
\usepackage[hidelinks]{hyperref}
% \usepackage[nameinlink,noabbrev]{cleveref}
```

Watch out. Package manuals and official journal templates override this general rule. If a class file gives instructions, follow them.

28. Common beginner mistakes in the preamble

Mistake	Better approach
Loading every package	Start small; activate modules only when needed.
Mixing bibliography systems	Choose `biblatex/biber` or `natbib/BibTeX`.
Mixing several font packages	Choose one coherent text/math font setup.
Loading `cleveref` before `hyperref`	Usually load `hyperref` first, then `cleveref`.
Forcing all floats with `[H]`	Prefer `[htbp]` and let LaTeX manage placement.
Manual formatting everywhere	Put global rules in the preamble once.
Hard-coding repeated notation	Define macros with `\newcommand` or `\DeclareMathOperator`.
One giant `main.tex`	Split sections and exported tables with `\input`.
Using `\\` for normal paragraph spacing	End a paragraph with a blank line; reserve `\\` for environments where a line/row break is meaningful.
Typing figure/table numbers manually	Use `\label` and `\ref` / `\cref`.
Using `$$ ... $$`	Prefer `\[ ... \]` for unnumbered display math or `equation/align` for numbered math.
Hard-coding figure widths in inches everywhere	Prefer proportions such as `width=.8\linewidth`.
Putting journal-specific formatting into every project	Keep a simple personal template and let official journal/class files control final styling.

29. Ready-to-copy quantitative homework preamble

This is a good default when you want something simple but useful.

```
\documentclass[12pt]{article}

% ---------- Math ----------
\usepackage{amsmath}
\usepackage{amssymb}
\usepackage{amsfonts}
```

```
\usepackage{mathtools}
\usepackage{bm}

% ---------- Layout ----------
\usepackage[margin=1in]{geometry}

% ---------- Tables ----------
\usepackage{booktabs}

% ---------- Links ----------
\usepackage[hidelinks]{hyperref}

% ---------- Common quantitative notation ----------
\newcommand{\E}{\mathbb{E}}
\newcommand{\Prob}{\mathbb{P}}
\newcommand{\R}{\mathbb{R}}
\DeclareMathOperator{\Var}{Var}
\DeclareMathOperator{\Cov}{Cov}
\DeclareMathOperator{\plim}{plim}
\DeclareMathOperator*{\argmax}{arg\,max}
\DeclareMathOperator*{\argmin}{arg\,min}

% ---------- Optional modules ----------
% \usepackage{graphicx}      % figures
% \usepackage{float}         % [H]
% \usepackage{siunitx}       % decimal-aligned tables
% \usepackage{setspace}      % line spacing
% \usepackage{fancyhdr}      % headers / footers

\title{ECONS XXX -- Homework X}
\author{Your Name}
\date{\today}

\begin{document}
\maketitle

% Start your homework here.

\end{document}
```

30. Ready-to-copy research-paper preamble

This adds the pieces commonly useful for an empirical paper while remaining manageable.

```
\documentclass[12pt]{article}

% ---------- Typography / layout ----------
\usepackage[margin=1in]{geometry}
\usepackage{microtype}

% ---------- Mathematics ----------
\usepackage{amsmath}
\usepackage{amssymb}
\usepackage{amsfonts}
\usepackage{mathtools}
\usepackage{bm}

% ---------- Figures / tables ----------
\usepackage{graphicx}
\usepackage{booktabs}
\usepackage{siunitx}
\usepackage{threeparttable}

% ---------- Optional paper tools ----------
% \usepackage{setspace}
% \onehalfspacing
% \usepackage{pdflscape}
% \usepackage[section]{placeins}

% ---------- Bibliography: choose ONE system ----------
% \usepackage[round,authoryear]{natbib}
```

```
%
% OR:
% \usepackage{csquotes}
% \usepackage[backend=biber,style=authoryear]{biblatex}
% \addbibresource{references.bib}

% ---------- Links / references ----------
\usepackage[hidelinks]{hyperref}
% \usepackage[nameinlink,noabbrev]{cleveref}

% ---------- Example quantitative notation ----------
\newcommand{\E}{\mathbb{E}}
\newcommand{\Prob}{\mathbb{P}}
\newcommand{\R}{\mathbb{R}}
\DeclareMathOperator{\Var}{Var}
\DeclareMathOperator{\Cov}{Cov}
\DeclareMathOperator{\Corr}{Corr}
\DeclareMathOperator{\plim}{plim}
\DeclareMathOperator*{\argmax}{arg\,max}
\DeclareMathOperator*{\argmin}{arg\,min}

% ---------- Project folders ----------
\graphicspath{{figures/}}

\title{Paper Title}
\author{Your Name}
\date{\today}

\begin{document}
\maketitle

% \input{sections/introduction}
% \input{sections/model}
% \input{sections/data}
% \input{sections/results}

\end{document}
```

31. Ready-to-copy theory-paper preamble

For a theory note or model-heavy field paper, theorem environments and notation consistency become more important than regression-table machinery.

```
\documentclass[12pt]{article}

% ---------- Layout / typography ----------
\usepackage[margin=1in]{geometry}
\usepackage{microtype}

% ---------- Math ----------
\usepackage{amsmath,amssymb,amsfonts,mathtools,bm}
\usepackage{amsthm}

% ---------- Theorem environments ----------
\newtheorem{theorem}{Theorem}[section]
\newtheorem{proposition}[theorem]{Proposition}
\newtheorem{lemma}[theorem]{Lemma}
\newtheorem{corollary}[theorem]{Corollary}
\theoremstyle{definition}
\newtheorem{assumption}[theorem]{Assumption}
\newtheorem{definition}[theorem]{Definition}
\theoremstyle{remark}
\newtheorem{remark}[theorem]{Remark}

% ---------- Notation ----------
\newcommand{\E}{\mathbb{E}}
\newcommand{\R}{\mathbb{R}}
\DeclareMathOperator*{\argmax}{arg\,max}
\DeclareMathOperator*{\argmin}{arg\,min}
\DeclarePairedDelimiter{\abs}{\lvert}{\rvert}
\DeclarePairedDelimiter{\norm}{\lVert}{\rVert}
```

```
% ---------- References ----------
\usepackage[hidelinks]{hyperref}
% \usepackage[nameinlink,noabbrev]{cleveref}

\title{Theory Paper Title}
\author{Your Name}
\date{\today}

\begin{document}
\maketitle

% \input{sections/model}
% \input{sections/equilibrium}
% \input{sections/proofs}

\end{document}
```

32. Module checklist: add only what the project needs

Need	Package / setup	Typical reason
Basic math	`amsmath, amssymb, mathtools`	Equations, alignments, symbols
Bold Greek/vector math	`bm`	$\boldsymbol{\beta}$, $\boldsymbol{x}$
Professional tables	`booktabs`	Clean regression/summary tables
Decimal alignment	`siunitx`	Numeric columns
Table notes	`threeparttable`	Notes under empirical tables
Figures	`graphicx`	Import PDF/PNG/JPG graphics
Figure panels	`subcaption`	Panel A / Panel B
Float barriers	`placeins`	Keep floats near a section
Landscape pages	`pdflscape`	Very wide tables
Theorems/proofs	`amsthm`	Theory papers
Line spacing	`setspace`	Referee/course requirements
Bibliography	`natbib` or `biblatex`	Citations and references
Smart refs	`cleveref` after `hyperref`	Automatic "Figure/Table/Equation" names
Draft TODOs	`todonotes`	Visible drafting reminders

33. My recommended master-template strategy

Keep one file with three layers:

1. **Always on:** core math, geometry, and one table package.
2. **Commented modules:** figures, bibliography, theorem tools, landscape pages, code, TODOs, special spacing.
3. **Your own macros:** notation that you repeatedly use in your field.

A compact master block might look like this:

```
% ==================== ALWAYS ON ====================
\usepackage{amsmath,amssymb,amsfonts,mathtools,bm}
\usepackage[margin=1in]{geometry}
\usepackage{booktabs}
\usepackage[hidelinks]{hyperref}

% ==================== OPTIONAL ====================
% \usepackage{graphicx}       % figures
% \usepackage{float}          % [H]
% \usepackage{subcaption}     % subfigures
% \usepackage{siunitx}        % aligned numbers
% \usepackage{threeparttable} % table notes
% \usepackage{setspace}       % 1.5 / double spacing
% \usepackage{fancyhdr}       % headers / footers
% \usepackage{amsthm}         % theorem environments
% \usepackage{pdflscape}      % landscape pages

% ==================== ECON MACROS ==================
```

```
% \newcommand{\E}{\mathbb{E}}
% \newcommand{\Prob}{\mathbb{P}}
% \DeclareMathOperator{\Var}{Var}
% \DeclareMathOperator{\Cov}{Cov}
% \DeclareMathOperator{\plim}{plim}
% \DeclareMathOperator*{\argmax}{arg\,max}
```